

Kumpel et al. (2024)

The diagram illustrates the metabolic pathways of sulfur in the periplasm, membrane, and cytoplasm. Key components and reactions include:

- Periplasm:**
 - Sox Pathway:** SoxB and SoxCD are involved in the oxidation of sulfur. SoxB converts H_2O and SO_4^{2-} to 2H^+ and $\text{S}_2\text{O}_3^{2-}$. SoxCD converts 6H^+ and 6e^- to $3\text{H}_2\text{O}$ and SO_4^{2-} . SoxYZ is a central protein that can be oxidized to $\text{S}_2\text{O}_3^{2-}$ or reduced to SO_4^{2-} .
 - Other Periplasmic Reactions:**
 - $2\text{S}_2\text{O}_3^{2-} \xrightarrow{\text{TsdA}} \text{S}_4\text{O}_6^{2-}$
 - $\text{SO}_3^{2-} \xrightarrow[\text{SorT}]{\text{SorAB}} \text{SO}_4^{2-}$
- Membrane:**
 - SoeABC and DoxDA:** These proteins are involved in the reduction of MQ to MQH_2 .
- Cytoplasm:**
 - Dsr and sHdr:** Dsr is involved in the reduction of HSO_3^- to SO_4^{2-} . sHdr is involved in the reduction of SO_4^{2-} to $\text{S}_2\text{O}_3^{2-}$ and $\text{S}_4\text{O}_6^{2-}$.

Legend:

type I		type II	
sHdrB1	sHdrB2	sHdrB1	sHdrB3
sHdrC1	sHdrC2	sHdrC1	sHdrA
sHdrA	sHdrA	sHdrA	sHdrA
sHdrH	sHdrI	sHdrH	sHdrH